

Recitation 9 — Answer Key

Q1) Answer Key

- (a) To find the YTM for each bond, we solve for r in the standard bond pricing equation. Expanding for $n = 5$ years:

Bond A (priced at \$980):

$$980 = \frac{60}{(1+r)} + \frac{60}{(1+r)^2} + \frac{60}{(1+r)^3} + \frac{60}{(1+r)^4} + \frac{60}{(1+r)^5} + \frac{1000}{(1+r)^5}$$

Bond B (priced at \$1,050):

$$1050 = \frac{60}{(1+r)} + \frac{60}{(1+r)^2} + \frac{60}{(1+r)^3} + \frac{60}{(1+r)^4} + \frac{60}{(1+r)^5} + \frac{1000}{(1+r)^5}$$

Solving numerically for r in each case:

- **Bond A** (priced at \$980): YTM $\approx 6.48\%$
- **Bond B** (priced at \$1,050): YTM $\approx 4.85\%$

The exact YTMs demonstrate that Bond A offers a higher return than Bond B for the same risk, indicating it is priced more attractively.

- (b) To exploit an arbitrage opportunity, an investor should **buy Bond A** (undervalued, yielding a higher return) and **sell Bond B** (overvalued, yielding a lower return). The assumption is that the market will eventually correct these discrepancies, and the prices of Bond A and Bond B will adjust such that their YTMs align, given their identical risk profiles and cash flows.
- (c) The relationship between bond price and yield illustrates the fundamental **inverse relationship**: as the price of a bond decreases, its yield increases, and vice versa. This is because the fixed coupon payments and maturity value represent a larger return percentage on a lower purchase price, and a smaller return percentage on a higher purchase price.

In this scenario, Bond A's lower price relative to Bond B results in a higher yield, demonstrating that as bond prices fall, yields rise. This principle is a fundamental concept in bond valuation, illustrating how returns adjust to reflect changes in bond pricing.

Q2) Answer Key

1. Effective Capital per Effective Worker Evolution

Begin with the standard capital accumulation equation:

$$K_{t+1} = (1 - \delta)K_t + sY_t$$

Define effective capital per effective worker as $k_t = K_t/(A_tL)$, so $K_t = A_tLk_t$. With labor-augmenting technology, rewrite the production function as:

$$Y_t = K_t^{0.5}(A_tL)^{0.5}$$

so that effective output per effective worker is:

$$y_t = \frac{Y_t}{A_tL} = \left(\frac{K_t}{A_tL} \right)^{0.5} = k_t^{0.5}$$

Dividing both sides of the capital accumulation equation by $A_{t+1}L = (1 + g)A_tL$:

$$\frac{K_{t+1}}{A_{t+1}L} = \frac{(1 - \delta)K_t}{(1 + g)A_tL} + \frac{sY_t}{(1 + g)A_tL}$$

The left-hand side is k_{t+1} . Substituting k_t and $y_t = k_t^{0.5}$ on the right-hand side yields the **law of motion**:

$$k_{t+1} = \frac{(1 - \delta)k_t + s k_t^{0.5}}{1 + g}$$

This equation shows that effective capital per effective worker next period depends on what survives depreciation plus new investment, both deflated by the growth factor $(1 + g)$ that continuously expands the effective labor denominator.

2. Steady-State Analysis

At the steady state, $k_{t+1} = k_t = k^*$ by definition. Substituting into the law of motion:

$$k^* = \frac{(1 - \delta)k^* + s k^{*0.5}}{1 + g}$$

Multiply both sides by $(1 + g)$:

$$(1 + g)k^* = (1 - \delta)k^* + s k^{*0.5}$$

Collect the k^* terms on the left:

$$(1 + g)k^* - (1 - \delta)k^* = s k^{*0.5}$$

$$(g + \delta) k^* = s k^{*0.5}$$

Divide both sides by $k^{*0.5}$ (valid since $k^* \neq 0$):

$$(g + \delta) k^{*0.5} = s$$

$$k^{*0.5} = \frac{s}{g + \delta}$$

Squaring both sides:

$$k^* = \left(\frac{s}{g + \delta} \right)^2$$

The steady-state level of effective capital per effective worker depends:

- **Positively** on the savings rate s — higher savings funds more investment, raising k^* .
- **Negatively** on the depreciation rate δ — faster depreciation erodes the capital stock.
- **Negatively** on the technological growth rate g — faster technological progress continuously dilutes effective capital by expanding $A_t L$, requiring more investment just to keep k_t constant.

3. Long-Run Growth Implications

With continuous technological progress, the economy does *not* reach a steady state in terms of actual capital per worker K/L or output per worker Y/L . Instead, it reaches a steady state only in *effective* units. The long-run implications are:

- (a) **Sustained per capita growth.** Even in the steady state (where k^* is constant), output per capita Y/L grows continuously at the rate of technological progress g , because A_t keeps rising.
- (b) **Total output growth.** If the labor force grows at rate n , total output grows at:

$$\text{Growth of } Y = g + n$$

reflecting both technological progress and labor force expansion.

- (c) **Role of technology.** Unlike changes in s or δ , which only affect the *level* of the balanced growth path, technological progress g drives the *rate* of long-run per capita growth. This is the central message of the Solow model: capital accumulation alone cannot sustain growth; only technological progress can.